

PAPER TITLE: Lagrangian Derivation under UQFF v5.78

Star-Magic UQFF
PAPER-NNNN / Session SESSION
DATE

Abstract

State the field / interaction whose Lagrangian is being derived, the canonical UQFF skeleton that hosts it, the Euler–Lagrange equation(s) obtained, and the observable they predict.

1 Field Content

- Scalar field(s): $\phi_{\text{SCm}}, \phi_{\text{UA}}$ – vacuum order parameters.
- Vector/tensor field(s): list, with mass dimension.
- Coupling constants: $\beta_i \approx 0.603$, $\kappa = 5 \times 10^{-4}/\text{day}$, $F_{\text{TRZ}} = 0.1$.

2 Lagrangian Density

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{kin}} + \mathcal{L}_{\text{grav}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{res}} - V(\rho_{\text{vac}}) \quad (1)$$

$$\mathcal{L}_{\text{buoy}} = \frac{1}{2} \rho_{\text{vac}} \frac{4\pi}{3} c^2 \phi^2 \cos(\pi t_n) \quad (2)$$

with $\rho_{\text{vac}} \in \{\rho_{\text{vac,SCm}}, \rho_{\text{vac,UA}}\}$ depending on the regime (structural G9 vs G7, see Sec. 5).

3 Euler–Lagrange Equation

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0 \implies \boxed{< EoMequationhere >} \quad (3)$$

4 Universal Inertia Invariant

The invariant differential arising from (2):

$$U_I = 3 \rho_{\text{vac}} \frac{4\pi}{3} c^2 \cos(\pi t_n) \quad (4)$$

Sign regimes: $U_I > 0$ centripetal (mass-building), $U_I = 0$ zero-point gravity ($t_n = \frac{1}{2}, \frac{3}{2}, \dots$), $U_I < 0$ centrifugal (negentropic void).

5 Observable Prediction & Closure

Quantity	Predicted (UQFF v5.78)	Observed (cite)
<code>jobservable_i</code>	<code>jvalue_i</code>	<code>jvalue_i [[?]]</code>

6 Cross-references

- Closed-form r_{hz} : `QCalcGeom.py::solve_habitable_zone` (v2.3.0).
- Universal Inertia: `QCalcGeom.py::universal_inertia`.
- Parent skeleton: `uqff_lagrangian_derivation.py`.

References

- [1] `jDOI/arXiv real citationi`.
- [2] PAPER_409 (Star-Magic): hydrogen geometry $\rho_{\text{vac,SCm}}$.
- [3] PAPER_1160 (Star-Magic): G7 $|SO(5)|$ closure.